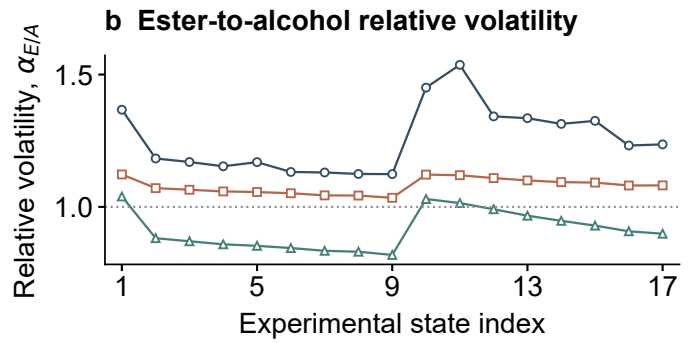
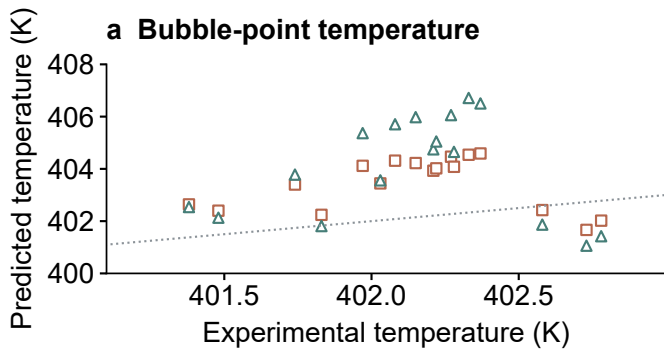


Extractive distillation: sec-butyl alcohol / sec-butyl acetate / DMF

101.3 kPa | 17 ternary VLE states | Equimolar alcohol-ester feed: 1 mol s⁻¹



c VLE prediction errors

Metric	ThermoFormer	UNIFAC
Vapor-composition MAE	0.017202	0.035060
Temperature MAE (K)	1.5323	2.3553
Max. temperature error (K)	2.2342	4.3833
Relative-volatility MAE	0.175320	0.341593
Volatility-direction agreement	17/17	3/17

E = ester; A = alcohol.

Composition MAE: 3 components, 17 states.

d Selected column configuration

Parameter	Value
DMF/feed mass ratio	32:1
DMF feed (mol s ⁻¹)	41.651671
Reflux ratio	48
Equilibrium stages	183
Alcohol-ester feed stage	159
DMF feed stage	16
Distillate flow (mol s ⁻¹)	0.492462
Bottoms flow (mol s ⁻¹)	42.159209
Internal vapor flow (mol s ⁻¹)	24.130653

e Design-search ranges

Calculation	Evaluated parameter sets
Initial search	Mass 2:1; N = {40, 80, 160}; R = {3, 5, 8, 12, 20, 40}; $n_S = \{4, 8, 16\}$; n_F near {0.35, 0.60, 0.85}N.
ThermoFormer refinement	Mass 2:1; N = {160, 300}; R = 4-10, step 0.5; $n_S = \{8, 20, 40\}$; $n_F = N - \{20, 40, 60\}$.
Solvent-ratio search	Mass m = {4, 8, 16, 32}; N = 160; $n_F = 140$; $n_S = \{12, 24, 36\}$; R = {m, 1.5m, 2m, 3m}.

Stage-boundary check Mass 32:1; R = 48; $n_S = 16$; $n_F = N - 24$.

f Stage boundary and common column

Model	N	n_F	Purity	Recovery	Met
ThermoFormer	182	158	99.499976	97.999977	No
ThermoFormer	183	159	99.508632	98.008502	Yes
UNIFAC	183	159	26.451879	26.053107	No

Targets: purity ≥ 99.5 mol%; recovery $\geq 98\%$.

Compliance uses unrounded values.

g Product compositions (mol%)

Stream	Componer	ThermoFormer	UNIFAC
Distillate	Alcohol	0.485730	73.540101
	Ester	99.508632	26.451879
	DMF	0.005638	0.008021
Bottoms	Alcohol	1.180307	0.326958
	Ester	0.023619	0.876996
	DMF	98.796075	98.796047

